WEEK 06 – FEATURE ENGINEERING

Moneyball: IPL Player Valuation | Team 05
Course: 24ADI204 – Data Science and Visualization

Team Members
Divyesh D S – 25BAD022 Kavi Prakash M – 25BAD043
Mithuin P S – 25BAD057 Madhan P M – 25BAD503
Renganathan G M – 25BAD087

1. Objective

Apply feature engineering to the Week 3 cleaned IPL dataset through descriptive statistics, skewness analysis and correction, numerical scaling, and categorical encoding.

2. Dataset Overview

The supplied dataset contains **3,311 rows** and **27 columns**, including **17 numerical** and **10 categorical** columns. It has **13,099 missing cells** and **0 duplicate rows**. The source CSV was not modified.

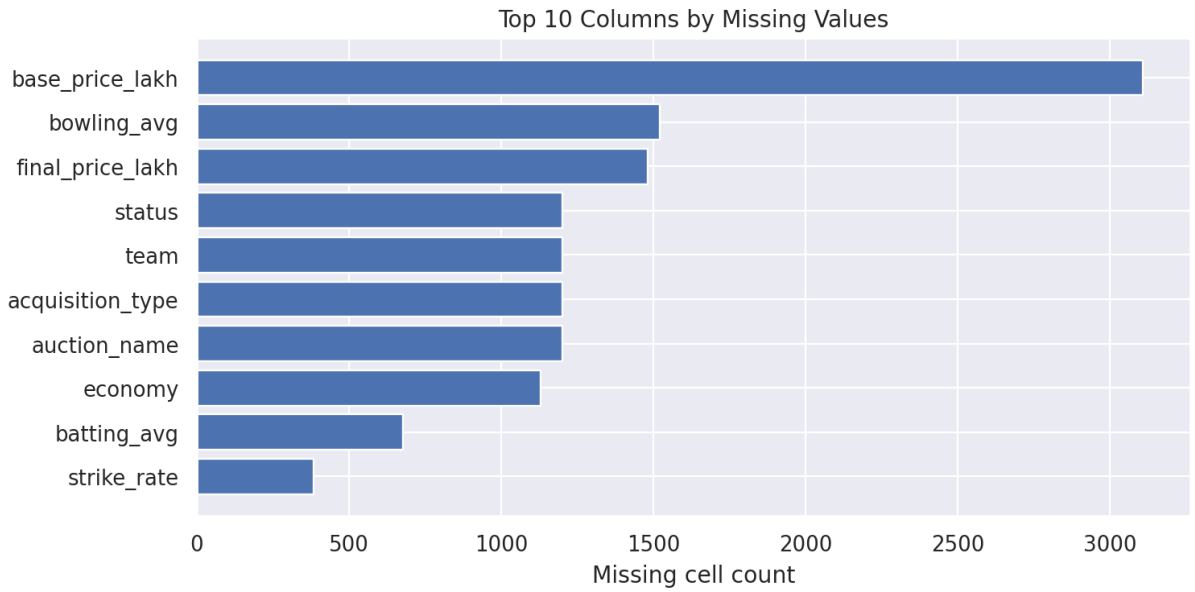


Figure 1. Ten columns with the highest missing-value counts.

3. Original Distributions

Histograms and density curves were plotted for runs, strike rate, batting average, wickets, economy and final auction price to inspect the numerical distributions.



Figure 2. Original distributions of selected IPL features.

4. Skewness Correction Using Yeo–Johnson

Features with absolute skewness greater than 1 were selected. The selected features were: runs, balls_faced, fours, sixes, balls_bowled, runs_conceded, wickets, final_price_lakh, batting_avg, economy, bowling_avg. Yeo–Johnson transformation was applied to their non-missing values using standardize=False. Missing values were not filled during this transformation.



Figure 3. Skewness before and after transformation.

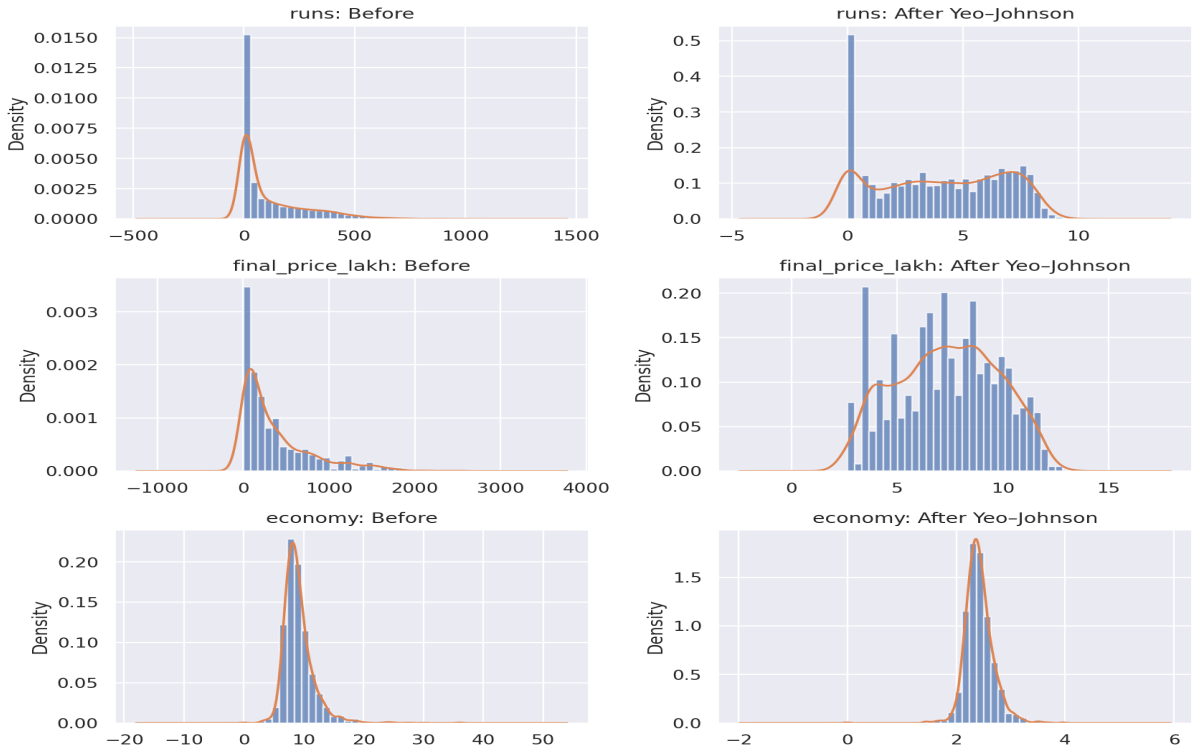


Figure 4. Distribution examples before and after transformation.

Feature	Before	After	Absolute skew reduced?
runs	1.630	-0.077	Yes
balls_faced	1.543	-0.078	Yes
fours	1.887	0.133	Yes
sixes	2.224	0.262	Yes

Feature	Before	After	Absolute skew reduced?
balls_bowled	1.209	-0.113	Yes
runs_conceded	1.119	-0.164	Yes
wickets	1.635	0.249	Yes
final_price_lakh	1.682	-0.048	Yes
batting_avg	1.186	-0.095	Yes
economy	2.718	0.036	Yes
bowling_avg	2.775	-0.037	Yes

5. Missing-Value Handling and Scaling

For scaling, missing numerical values were imputed using the median strategy. Missing cells in the numeric scaling matrix after imputation: **0**. Min-Max scaling maps each numerical feature to the 0–1 range. Standard scaling produces approximately zero mean and unit population standard deviation.

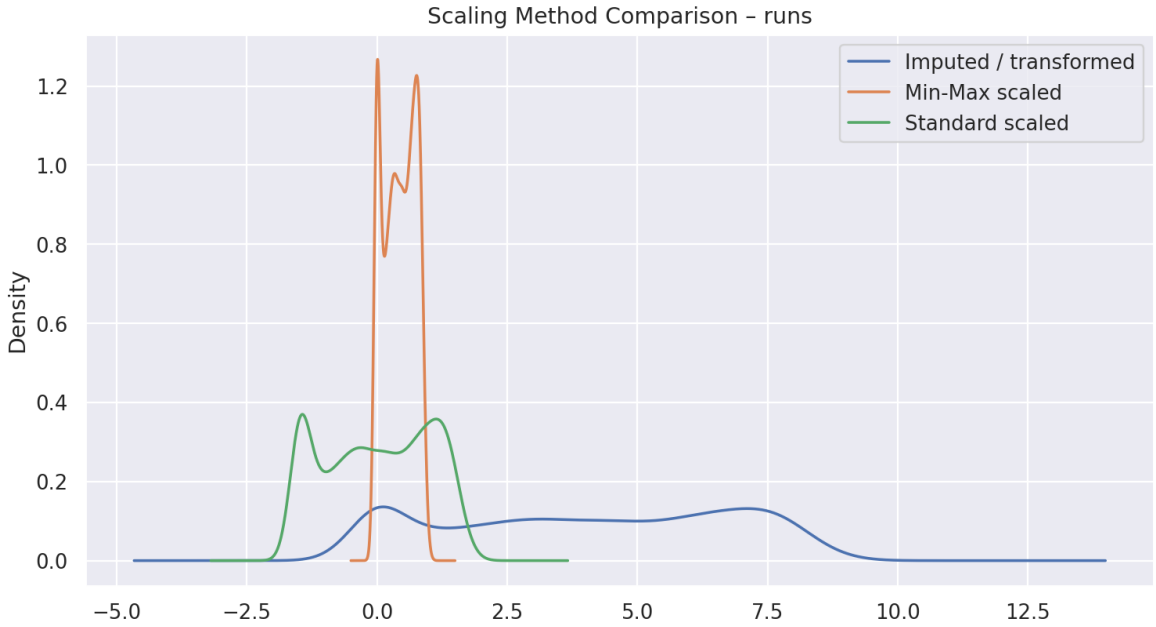


Figure 5. Distribution comparison for runs.

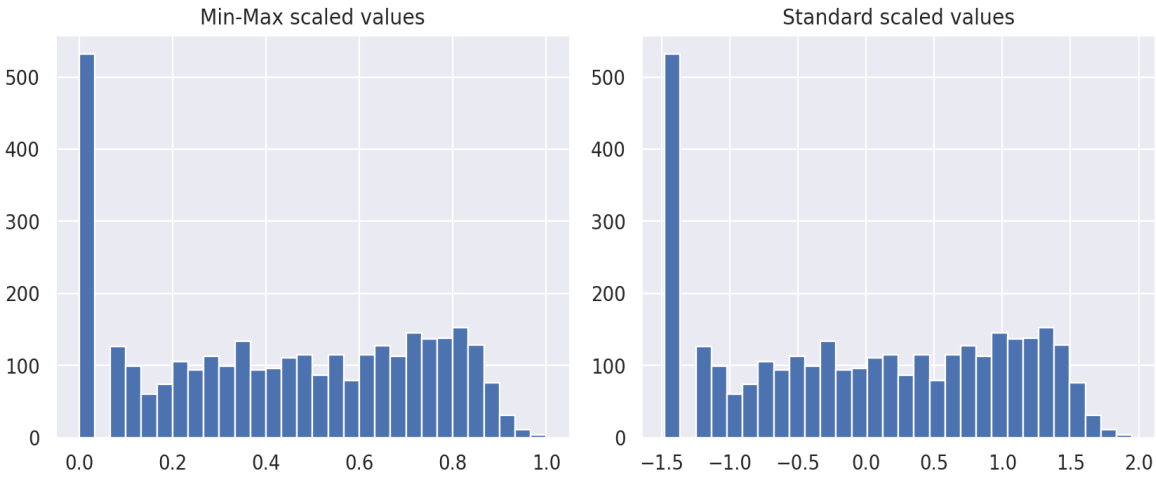


Figure 6. Min-Max and Standard scaled values for runs.

6. Categorical Encoding

One-Hot Encoding was applied to role, nationality, bowl_style, bat_style. These columns produced **35 encoded columns**. Label Encoding was applied to 'role' and generated 6 category codes. These codes are identifiers and do not represent an ordinal ranking.

Category	Encoded value
All-Rounder	0
Batsman	1
Batter	2
Bowler	3
Unknown	4
Wicket-Keeper	5

7. Key Findings

• 11 numerical features met the |skewness| > 1 selection rule.
• Absolute skewness decreased for 11 of the 11 selected features.
• Median imputation removed missing values from the numeric scaling matrix.
• Both scaling methods were demonstrated and compared visually.
• One-Hot Encoding expanded the four selected categorical fields into 35 indicator columns.

8. Conclusion

Week 6 feature engineering was completed on the supplied IPL dataset. The analysis identified highly skewed features, applied Yeo–Johnson transformation, imputed numeric missing values for scaling, compared Min-Max and Standard scaling, and encoded categorical variables using One-Hot and Label Encoding. The original cleaned CSV remains unchanged.

Note: The notebook creates separate outputs for skewness correction, numerical scaling and categorical encoding. For predictive modelling, fit transformations on training data only and apply them to validation/test data to avoid data leakage.

9. Output Dimensions

Skewness-corrected full dataset: 3311 × 27
Min-Max scaled numeric matrix: 3311 × 17
Standard scaled numeric matrix: 3311 × 17
One-Hot encoded categorical matrix: 3311 × 35
Label encoded output: 3311 rows × 2 columns